

## CSC 453: Advanced Programming Language

### 2<sup>rd</sup> Semester \_2025/2026 Term Project

#### Objective(s):

1. An ability to design and test solutions to programming problems implemented with various programming paradigms.
2. An ability to discuss the strengths and limitations of different programming paradigms in solving programming problems.
3. An ability to evaluate the programming paradigm of modern programming languages.

**Tool(s)/Software:** Open (A **new** language must be used).

#### Groups work:

- The project is a teamwork.
- Each team should consist of 4 to 5 students including the team leader.
- All your submissions must include your academic information (Name, ID, Group Name).
- Each member of the team must participate equally in the project. Otherwise, if a student didn't participate well in its team's project marks will be deducted according to the inactive student's participation.
- The team leader is responsible fully of dividing the work equally between members and reporting any inactive student.
- Your work should present the effort of 4-5 students.

#### Submitting Deadlines:

- **Sunday 08/02/2026:** Defining the project ideas, setting up the team leader, and defining the objectives of choosing this idea to be implemented (**1 Mark**).  
Discussion board.
- **Tuesday 31/03/2026:** Submitting a COMPREHENSIVE proposal with an explanation about the scope and objectives along with full clarification of your project and the timeline of completing the project (**3 Mark**).
- **Saturday 09/05/2026:** Submitting Project Report + Implementation. (**10 Marks**)
- **Sunday 31/05/2026:** Project Presentation. (**1 Marks**)

### Project Specifications:

Choose any project from the given list (or you may propose your own) and you need to implement it in two languages, one from your curriculum (Java, C++, PHP, Android etc.) and the second language should be a **new** language from the industry. Like *Python* or any web-based language (*ASP.net*) and smartphone/desktop language etc.

Build a server-client application for any of the following (as an example):

- a. Sale management and inventory system
- b. Weather Monitoring System
- c. Students' management and Attendance system
- d. Employee management and Payroll system
- e. Stock Trading Simulation

**Note:** you can choose any other server-client application if you want.

Students are expected to demonstrate skills in the following areas:

**1. Design solutions to programming problems implemented with various programming paradigms (6 marks):**

Implement fully functional application in two languages from different paradigm, the two applications should work correctly without errors, validate user entries and include complete functionalities (insert/ update/ delete).

**2. Compare the strengths and limitations of different programming paradigms in solving programming problems (2 marks):**

Compare between the two programming languages in term of the Syntax, Dependencies, Supported Paradigm, and Memory Management

**3. Evaluate the programming paradigm of modern programming languages (2 marks):**

Compare the implementation of the system functions in both selected programming languages (minimum 4 Functions) (compare the codes and syntax)

**Note:**

- The selected languages should be from different paradigms.
- At least one new language should be used for implementation.
- The comparison between the selected languages should be included in the final report.

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**Final report should contain:**

- The role of each student in the project.
- The report contains the updated timeline.
- The implementation of the project.
- The comparison between the selected languages.
- The difficulties that the team or an individual may face when implementing the proposal.
- The improvement that could be made if more time has been given.